Approximation Methods Reading

We saw how to use the characteristic polynomial to find the eigenvalues (and then the eigenvectors) for matrices. Such methods work fine for the small matrices considered there, but they are not practical for many larger matrices.

Since large matrices turn up in all kinds of applications, we need another way to find eigenvalues and eigenvectors. Several related approaches for dealing with large matrices are described in this lesson. All have their basis in the Power Method, so we start with that.

The Power Method

The Power Method is an iterative algorithm that gets its name from how it is implemented. Given a square matrix A, we start with a fixed vector $\mathbf{x}_0$ and compute the sequence $\mathbf{x}_0, A\mathbf{x}_0, A^2\mathbf{x}_0, A^3\mathbf{x}_0, \cdots$. Remarkably, in many cases the resulting sequence of vectors will approach an eigenvector of A. We begin with an example and leave the discussion of when and why the Power Method works for later in this lesson.

Suppose that

$$A = \begin{bmatrix} 1 & 3 \\ 2 & 2 \end{bmatrix}, \quad \mathbf{x}_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \tag{1}$$

and let

$$\mathbf{x}_1 = A\mathbf{x}_0, \quad \mathbf{x}_2 = A\mathbf{x}_1 = A^2\mathbf{x}_0, \quad \mathbf{x}_3 = A\mathbf{x}_2 = A^3\mathbf{x}_0, \cdots$$

Table 1 gives $\mathbf{x}_k$ to $\mathbf{x}_7$ for A and $\mathbf{x}_0$ above.

Table 1 $\mathbf{x}_k = A^k\mathbf{x}_0$ for $k = 0, 1, \ldots, 7$

k	0	1	2	3	4	5	6	7
$\mathbf{x}_k$	$\begin{bmatrix} 1 \\ 0 \end{bmatrix}$	$\begin{bmatrix} 1 \\ 2 \end{bmatrix}$	$\begin{bmatrix} 7 \\ 6 \end{bmatrix}$	$\begin{bmatrix} 25 \\ 26 \end{bmatrix}$	$\begin{bmatrix} 103 \\ 102 \end{bmatrix}$	$\begin{bmatrix} 409 \\ 410 \end{bmatrix}$	$\begin{bmatrix} 1639 \\ 1638 \end{bmatrix}$	$\begin{bmatrix} 6553 \\ 6554 \end{bmatrix}$

Often the components of $\mathbf{x}_k$ grow with k when forming this type of sequence. Since scalar multiples of eigenvectors are still eigenvectors, we control the size of each vector in the sequence by scaling.

DEFINITION 16

THE POWER METHOD: For each $k \geq 0$,

(a) Let λ_k denote the largest component (in absolute value) of $A\mathbf{x}_k$. (We call λ_k a **scaling factor**.)

(b) Set $\mathbf{x}_{k+1} = \frac{1}{\lambda_k} A\mathbf{x}_k$.

Repeat (a) and (b) to generate the sequence $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \ldots$.

Scaling in this way ensures that the largest component (in absolute value) of each vector is either 1 or −1. For example, starting with $\mathbf{x}_1$ and A in (1), we compute $\mathbf{x}_2$ by

$$A\mathbf{x}_1 = \begin{bmatrix} 1 & 3 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} 0.5 \\ 1 \end{bmatrix} = \begin{bmatrix} 3.5 \\ 3 \end{bmatrix} \Rightarrow \lambda_1 = 3.5 \Rightarrow \mathbf{x}_2 = \frac{1}{3.5} A\mathbf{x}_1 = \begin{bmatrix} 1 \\ 0.857 \end{bmatrix}$$

We then compute $\mathbf{x}_3$ by

$$A\mathbf{x}_2 = \begin{bmatrix} 1 & 3 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \end{bmatrix} \Rightarrow \lambda_2 = 2 \Rightarrow \mathbf{x}_3 = \frac{1}{2} A\mathbf{x}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Table 2 shows $\mathbf{x}_k$ to $\mathbf{x}_7$ together with the scaling factors.

Table 2 The Power Method Applied to A and $\mathbf{x}_0$ in (1)

k	0	1	2	3	4	5	6	7
$\mathbf{x}_k$	$\begin{bmatrix} 1 \\ 0 \end{bmatrix}$	$\begin{bmatrix} 0.5 \\ 1 \end{bmatrix}$	$\begin{bmatrix} 1.000 \\ 0.8571 \end{bmatrix}$	$\begin{bmatrix} 0.9615 \\ 1.000 \end{bmatrix}$	$\begin{bmatrix} 1.000 \\ 0.9903 \end{bmatrix}$	$\begin{bmatrix} 0.9976 \\ 1.000 \end{bmatrix}$	$\begin{bmatrix} 1.000 \\ 0.9994 \end{bmatrix}$	$\begin{bmatrix} 0.9998 \\ 1.000 \end{bmatrix}$
σ_k	2	3.5	3.714	3.962	3.981	3.998	3.999	4.000

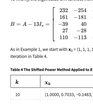


Figure 1 Vectors $\mathbf{x}_0, \ldots, \mathbf{x}_7$ from Table 2.

The entries in Table 2 and Figure 1 suggest that the sequence of vectors is getting closer and closer to $\mathbf{u} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

Computing $A\mathbf{u}$, we find that

$$A\mathbf{u} = \begin{bmatrix} 1 & 3 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \end{bmatrix} = 4\mathbf{u} \tag{2}$$

Hence $\mathbf{u} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ is an eigenvector of A with associated eigenvalue of $\lambda = 4$.

In Table 2, not only does the sequence of vectors converge to an eigenvector, but the sequence of scaling factors converges to the associated eigenvalue. This latter observation makes sense, because our scaling rule $\lambda_{k+1} = \frac{1}{\lambda_k} A\mathbf{x}_k$ can be expressed as

$$A\mathbf{x}_k = \sigma_k \mathbf{x}_{k+1}$$

Since the vectors are getting closer to an eigenvector (and each other), it follows that σ_k must be getting closer to an eigenvalue.

Typically, any nonzero $\mathbf{x}_0$ can be used as an initial vector for the Power Method. (However, see the note in the Computational Comments at the end of the lesson.) If an approximate value of an eigenvector is known, then using it for $\mathbf{x}_0$ can speed convergence.

Example 1

Find an eigenvector and associated eigenvalue for the matrix

$$A = \begin{bmatrix} 245 & -254 & -252 & -46 & -224 \\ 161 & -168 & -174 & -32 & -148 \\ -39 & 40 & 45 & 7 & 36 \\ 27 & -28 & -32 & -6 & -26 \\ 110 & -113 & -110 & -21 & -101 \end{bmatrix} \tag{3}$$

The matrix A is from (4) in Lesson 1.

Solution

We apply the Power Method starting with $\mathbf{x}_0 = (1, 1, 1, 1, 1)$. Table 3 gives the vectors $\mathbf{x}_1, \ldots, \mathbf{x}_9$.

To save space, in this lesson we sometimes write vectors horizontally instead of vertically.

Table 3 The Power Method Applied to A in Example 1

k	$\mathbf{x}_k$	σ_k
1	(1.0000, 0.4798, −0.1714, 0.1124, 0.4426)	19.74
2	(1.0000, 0.4693, −0.1717, 0.1124, 0.4431)	13.83
3	(1.0000, 0.4688, −0.1683, 0.1128, 0.4438)	12.94
4	(1.0000, 0.4674, −0.1678, 0.1118, 0.4442)	13.07
5	(1.0000, 0.4671, −0.1670, 0.1114, 0.4443)	13.01
6	(1.0000, 0.4668, −0.1668, 0.1112, 0.4444)	13.01
7	(1.0000, 0.4668, −0.1667, 0.1112, 0.4444)	13.00
8	(1.0000, 0.4667, −0.1667, 0.1111, 0.4444)	13.00
9	(1.0000, 0.4667, −0.1667, 0.1111, 0.4444)	13.00

The sequence settles to the vector (1.0000, 0.4667, −0.1111, 0.4444), and the sequence of scaling factors to 13.00. The components of the vector are recognizable as decimal approximations to rationals. Changing the decimals to equivalent rationals and multiplying by 18 to eliminate the fractions gives us

$$\mathbf{u} = \begin{bmatrix} 1.0000 \\ 0.6667 \\ -0.1667 \\ 0.1111 \\ 0.4444 \end{bmatrix} \approx \begin{bmatrix} 1 \\ 2/3 \\ -1/6 \\ 1/9 \\ 4/9 \end{bmatrix} \Rightarrow 18\,\mathbf{u} = \begin{bmatrix} 18 \\ 12 \\ -3 \\ 2 \\ 8 \end{bmatrix}$$

We express the entries as fractions and multiply by 18 to make it easier to check our answer. This would not be done in applications. Instead, typically we would compute enough iterations to achieve a desired accuracy and take the last vector in the sequence as the eigenvector.

To test our answer, we calculate

$$A\mathbf{u} = \begin{bmatrix} 245 & -254 & -252 & -46 & -224 \\ 161 & -168 & -174 & -32 & -148 \\ -39 & 40 & 45 & 7 & 36 \\ 27 & -28 & -32 & -6 & -26 \\ 110 & -113 & -110 & -21 & -101 \end{bmatrix} \begin{bmatrix} 18 \\ 12 \\ -3 \\ 2 \\ 8 \end{bmatrix} = \begin{bmatrix} 234 \\ 156 \\ -39 \\ 26 \\ 104 \end{bmatrix} = 13 \begin{bmatrix} 18 \\ 12 \\ -3 \\ 2 \\ 8 \end{bmatrix}$$

confirming that $\mathbf{u}$ is an eigenvector and that $\lambda = 13$ is the associated eigenvalue.

The Power Method will frequently find an eigenvalue and associated eigenvector. But which ones? To find out, first suppose that a matrix A has eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$ such that

$$|\lambda_1| > |\lambda_2| \geq |\lambda_3| \geq \cdots \geq |\lambda_n|$$

In this case, λ_1 is called the dominant eigenvalue of A.

The next theorem tells when the Power Method will converge to an eigenvector associated with the dominant eigenvalue. The proof of this theorem is given at the end of the lesson.

THEOREM 17

Let A be an $n \times n$ matrix with linearly independent eigenvectors $\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n$ and associated eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$, where λ_1 is dominant. Suppose that

$$\mathbf{x}_0 = c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + \cdots + c_n\mathbf{u}_n$$

where $c_1 \neq 0$. Then multiples of $A\mathbf{x}_0, A^2\mathbf{x}_0, \ldots$ converge to a scalar multiple of the eigenvector $\mathbf{u}_1$.

The Shifted Power Method

The Power Method can be used to find eigenvectors other than those associated with the dominant eigenvalue. Suppose that λ_1 is the dominant eigenvalue of A, and then let I be the dominant eigenvalue of $A - \lambda_1 I$, with associated eigenvector $\mathbf{u}_1$. Then

$$(A - \lambda_1 I)\mathbf{u}_1 = \lambda\mathbf{u}_1 \quad \Rightarrow \quad A\mathbf{u}_1 = (\lambda + \lambda_1)\mathbf{u}_1$$

Thus, if $\lambda_1 + \lambda = \lambda_2$, then λ_2 is an eigenvalue of A. Furthermore, since λ is the dominant eigenvalue of $A - \lambda_1 I$, and $\lambda = \lambda_2 - \lambda_1$, it follows that λ_2 is the eigenvalue of A that is farthest from λ_1. (Why?) Hence applying the Power Method to $A - \lambda_1 I$, will produce another eigenvalue and eigenvector of A. This is called the Shifted Power Method and is illustrated in the next example.

Example 2

For the matrix A given in Example 1, find the eigenvalue (and an associated eigenvector) farthest from $\lambda_1 = 13$.

Solution

To find λ_2, the eigenvalue of A farthest from $\lambda_1 = 13$, we first apply the Power Method to the matrix

$$B = A - 13I_5 = \begin{bmatrix} 232 & -254 & -252 & -46 & -224 \\ 161 & -181 & -174 & -32 & -148 \\ -39 & 40 & 32 & 7 & 36 \\ 27 & -28 & -32 & -19 & -26 \\ 110 & -113 & -110 & -21 & -114 \end{bmatrix}$$

As in Example 1, we start with $\mathbf{x}_0 = (1, 1, 1, 1, 1)$. In this case, the convergence is slower, so we report only every 10th iteration in Table 4.

Table 4 The Shifted Power Method Applied to $B = A - 13I_5$

k	$\mathbf{x}_k$	σ_k
10	(1.0000, 0.7033, −0.1483, 0.1483, 0.4450)	−13.77
20	(1.0000, 0.7078, −0.1461, 0.1461, 0.4383)	−13.86
30	(1.0000, 0.7107, −0.1447, 0.1447, 0.4340)	−13.92
40	(1.0000, 0.7123, −0.1438, 0.1438, 0.4315)	−13.96
50	(1.0000, 0.7132, −0.1434, 0.1434, 0.4302)	−13.98
60	(1.0000, 0.7137, −0.1431, 0.1431, 0.4294)	−13.99
70	(1.0000, 0.7140, −0.1430, 0.1430, 0.4290)	−13.99
80	(1.0000, 0.7141, −0.1429, 0.1429, 0.4288)	−16.00
90	(1.0000, 0.7142, −0.1429, 0.1429, 0.4287)	−16.00
100	(1.0000, 0.7142, −0.1429, 0.1429, 0.4286)	−16.00

We see that $\sigma_k \to -16$, which implies that $\lambda \to -16$ is an eigenvalue for B. Therefore $\lambda_2 = -16 + 13 = -3$ is the eigenvalue of A farthest from $\lambda_1 = 13$. Table 4 shows $\mathbf{u} \approx (1.000, 0.7142, -0.1429, 0.1429, 0.4286)$ is the associated eigenvector. We check this by computing

$$A\mathbf{u} = \begin{bmatrix} 245 & -254 & -252 & -46 & -224 \\ 161 & -166 & -174 & -32 & -148 \\ -39 & 40 & 45 & 7 & 36 \\ 27 & -28 & -32 & -6 & -26 \\ 110 & -113 & -110 & -21 & -101 \end{bmatrix} \begin{bmatrix} 1.0000 \\ 0.7142 \\ -0.1429 \\ 0.1429 \\ 0.4286 \end{bmatrix} \approx \begin{bmatrix} -2.9758 \\ 2.1286 \\ -0.4246 \\ -0.4256 \\ -1.2751 \end{bmatrix} \approx -3 \begin{bmatrix} 0.9919 \\ 0.7099 \\ -0.1415 \\ 0.1419 \\ 0.4250 \end{bmatrix} \approx -3\mathbf{u}$$

The approximations are a bit rough, but they can be refined by carrying more decimal places and computing additional iterations (see following note).

Carrying more decimal places in Example 2 gives (after 250 iterations)

$$\mathbf{u} = \begin{bmatrix} 1.00000000 \\ 0.71428571 \\ -0.14285714 \\ 0.14285714 \\ 0.42857143 \end{bmatrix}$$

Then

$$A\mathbf{u} = \begin{bmatrix} -3.00000000 \\ 2.14285714 \\ 0.42857143 \\ -0.42857143 \\ -1.28571428 \end{bmatrix} = -3 \begin{bmatrix} 1.00000000 \\ 0.71428571 \\ -0.14285714 \\ 0.14285714 \\ 0.42857143 \end{bmatrix} = -3\mathbf{u}$$

The Inverse Power Method

For the matrix A in Examples 1 and 2, we have found eigenvalues $\lambda_1 = 13$ and $\lambda_2 = -3$, and we also know that these are the positive and negative eigenvalues farthest from 0. Here we develop a method for finding the eigenvalue λ_3 that is *closest* to 0. If A is an invertible matrix with eigenvalue λ and associated eigenvector $\mathbf{x}$, then λ^{-1} is an eigenvalue of A^{-1} with associated eigenvector $\mathbf{x}$. Thus, in particular, if λ is the largest (in absolute value) eigenvalue of A^{-1}, then λ^{-1} must be the *smallest* (in absolute value) eigenvalue of A. (Why?) Therefore, applying the Power Method to A^{-1} will yield the smallest eigenvalue (and corresponding eigenvector) of A.

Example 3

For the matrix A in Example 1, find the smallest eigenvalue (in absolute value) and an associated eigenvector.

Solution

Here we apply the Power Method to

$$A^{-1} = \frac{1}{65} \begin{bmatrix} -646 & 698 & -420 & -1,430 & 620 \\ -481 & 487 & -306 & -995 & 448 \\ 99 & -112 & 70 & 221 & -96 \\ -27 & 80 & -64 & -143 & 54 \\ -290 & 329 & -176 & -663 & 264 \end{bmatrix}$$

generating a sequence of vectors of the form

$$\mathbf{x}_{k+1} = \frac{1}{\lambda_k} A^{-1}\mathbf{x}_k \tag{4}$$

For large matrices it can be difficult to compute A^{-1} accurately due to round-off error. This can be avoided by multiplying each side of (4) by A, yielding the sequence of linear systems

$$A\mathbf{x}_{k+1} = \frac{1}{\lambda_k}\mathbf{x}_k$$

Applying LU factorization (or a related method) to A can greatly improve computational efficiency when solving these systems.

For a change of pace (and another reason to be discussed later), let's take the initial vector to be $\mathbf{x}_0 = (1, 2, 3, 4, 5)$. Table 5 gives the results of every other iteration.

Table 5 The Inverse Power Method Applied to A

k	$\mathbf{x}_k$	σ_k
2	(1.0000, 0.6842, −0.1565, 0.1178, 0.4734)	0.2866
4	(1.0000, 0.6887, −0.1556, 0.0921, 0.4669)	0.7421
6	(1.0000, 0.6912, −0.1544, 0.0811, 0.4632)	0.9297
8	(1.0000, 0.6919, −0.1540, 0.0780, 0.4620)	0.9813
10	(1.0000, 0.6922, −0.1539, 0.0772, 0.4616)	0.9956
12	(1.0000, 0.6923, −0.1539, 0.0770, 0.4616)	0.9989
14	(1.0000, 0.6923, −0.1538, 0.0769, 0.4615)	0.9997
16	(1.0000, 0.6923, −0.1538, 0.0769, 0.4615)	0.9999

The output suggests that A^{-1} has eigenvalue $\lambda = 1$, so that $\lambda_3 = \lambda^{-1} = 1$ is an eigenvalue of A. We check the vector $\mathbf{w} = (1.0000, 0.6923, -0.1538, 0.0769, 0.4615)$ from Table 5 by computing

$$A\mathbf{w} = \begin{bmatrix} 245 & -254 & -252 & -46 & -224 \\ 161 & -168 & -174 & -32 & -148 \\ -39 & 40 & 45 & 7 & 36 \\ 27 & -28 & -32 & -6 & -26 \\ 110 & -113 & -110 & -21 & -101 \end{bmatrix} \begin{bmatrix} 1.0000 \\ 0.6923 \\ -0.1538 \\ 0.0769 \\ 0.4615 \end{bmatrix} \approx \begin{bmatrix} 1.0000 \\ 0.6920 \\ -0.1537 \\ 0.0769 \\ 0.4617 \end{bmatrix} \approx \mathbf{w}$$

Thus $\mathbf{w}$ is an eigenvector associated with eigenvalue $\lambda_3 = 1$.

The Shifted Inverse Power Method

So far we have found three eigenvalues (and associated eigenvectors) for A,

- The largest (in absolute value) λ_1,
- The eigenvalue λ_2 that is farthest from λ_1,
- The eigenvalue λ_3 that is closest to the origin.

Another way to look for eigenvalues is to start with the matrix $B = A - cI_5$ for some scalar c. Applying the Inverse Power Method to B will find the eigenvalue λ of B that is closest to the origin. Since $B = A - cI_5$, $\lambda + c$ is the eigenvalue of A that is closest to c.

Example 4

For the matrix A given in Example 1, find the eigenvalue that is closest to $c = 4$.

Solution

We start by setting

$$B = A - 4I_5 = \begin{bmatrix} 241 & -254 & -252 & -46 & -224 \\ 161 & -172 & -174 & -32 & -148 \\ -39 & 40 & 41 & 7 & 36 \\ 27 & -28 & -32 & -10 & -26 \\ 110 & -113 & -110 & -21 & -105 \end{bmatrix}$$

Now we apply the Inverse Power Method to B, starting out with the vector $\mathbf{x}_0 = (5, 4, 3, 2, 1)$. Table 6 includes every fifth iteration.

Table 6 The Shifted Inverse Power Method (Example 4)

k	$\mathbf{x}_k$	σ_k
5	(1.0000, 0.5363, −0.0292, 0.0141, 0.4931)	0.9416
10	(1.0000, 0.4936, 0.0051, −0.0026, 0.5013)	0.4616
15	(1.0000, 0.5008, −0.0008, 0.0003, 0.4998)	0.5053
20	(1.0000, 0.4999, 0.0001, 0.0000, 0.5000)	0.4993
25	(1.0000, 0.5000, 0.0000, 0.0000, 0.5000)	0.5001
30	(1.0000, 0.5000, 0.0000, 0.0000, 0.5000)	0.5000
35	(1.0000, 0.5000, 0.0000, 0.0000, 0.5000)	0.5000

We can see that $\lambda = 0.5$ is an eigenvalue for B^{-1}, so that $\lambda = 2$ is an eigenvalue for B. By shifting back, we find that $\lambda_4 = 2 + 4 = 6$ is an eigenvalue for A, with associated eigenvector $\mathbf{v} \approx (1, 0.5, 0, 0, 0.5)$. We check this by computing

$$A\mathbf{u} = \begin{bmatrix} 245 & -254 & -252 & -46 & -224 \\ 161 & -166 & -174 & -32 & -148 \\ -39 & 40 & 45 & 7 & 36 \\ 27 & -28 & -32 & -6 & -26 \\ 110 & -113 & -110 & -21 & -101 \end{bmatrix} \begin{bmatrix} 1.0 \\ 0.5 \\ 0 \\ 0 \\ 0.5 \end{bmatrix} = \begin{bmatrix} 6 \\ 3 \\ 0 \\ 0 \\ 3 \end{bmatrix} = 6\mathbf{u}$$

Applying the Shifted Inverse Power Method with different choices of c can yield other eigenvalues and eigenvectors. It is typically not efficient to do this for random values of c, but in some applications rough estimates of the eigenvalues are known. In these instances, the Shifted Inverse Power Method can turn estimates into accurate approximations.

COMPUTATIONAL COMMENTS

Some general remarks about the Power Method and related techniques discussed in this lesson:

- Similar to the approximation methods for finding solutions to linear systems, the Power Method is not overly sensitive to round-off error. In fact, each successive vector can be viewed as a starting point for the algorithm, so even if an error occurs, it typically will be corrected.
- The rate of convergence has differed in the examples we have considered. With the Power Method and its relatives, the larger the dominant eigenvalue is relative to the other eigenvalues, the faster the rate of convergence.
- The Power Method is guaranteed to work only on matrices whose eigenvectors span $\mathbb{R}^n$. However, in practice this method often also will work on other matrices, although convergence may be slower.
- Earlier we stated that the choice of starting vector $\mathbf{x}_0$ does not matter, and in virtually all cases this will be true. However it is possible to get misleading results from an unlucky choice of $\mathbf{x}_0$. For instance, in Example 3 we used $\mathbf{x}_0 = (1, 2, 3, 4, 5)$ in place of $(1, 1, 1, 1, 1)$, claiming at the time that this was done for "a change of pace." However, there was another reason. Table 7 gives the results when starting with $\mathbf{x}_0 = (1, 1, 1, 1, 1)$.

Table 7 New Application of the Inverse Power Method

k	σ_k	$\mathbf{x}_k$
5	−0.4059	(1.0000, 0.6758, −0.1624, 0.1624, 0.4866)
10	−0.4928	(1.0000, 0.6681, −0.1660, 0.1660, 0.4979)
15	−0.4990	(1.0000, 0.6669, −0.1666, 0.1666, 0.4997)
20	−0.4999	(1.0000, 0.6667, −0.1667, 0.1667, 0.5000)
25	−0.5000	(1.0000, 0.6667, −0.1667, 0.1667, 0.5000)
30	−0.5000	(1.0000, 0.6667, −0.1667, 0.1667, 0.5000)

This gives us $\lambda = -2$, which is an eigenvalue of A but not the eigenvalue closest to 0. We did not find $\lambda = 1$ because this choice of $\mathbf{x}_0$ happens to be in the span of the eigenvectors not associated with $\lambda = 1$. If we look at the statement of Theorem 17 again, we see that one of the conditions is violated.

In most applications, the entries of a matrix A are decimals and are subject to some degree of rounding. The likelihood of such an unlucky choice of $\mathbf{x}_0$ happening in practice is small.

- In cases where the eigenvalues satisfy
  $$|\lambda_1| = |\lambda_2| = \cdots = |\lambda_k| > |\lambda_{k+1}| \geq \cdots \geq |\lambda_n|$$
  and $\lambda_1 = \lambda_2 = \cdots = \lambda_k$, the Power Method will still work fine. However, if (for example) instead $\lambda_1 = -\lambda_2$, then the Power Method can produce strange results.

Proof of Theorem 17

Proof

To understand why the Power Method works, suppose that we have an $n \times n$ matrix A that has eigenvalues $\lambda_1, \ldots, \lambda_n$ such that

$$|\lambda_1| > |\lambda_2| \geq \cdots \geq |\lambda_n| \tag{5}$$

Assume that the associated eigenvectors $\mathbf{u}_1, \ldots, \mathbf{u}_n$ form a basis for $\mathbb{R}^n$. If $\mathbf{x}_0$ is an arbitrary vector, then there exist scalars $c_1, \ldots, c_n$ such that

$$\mathbf{x}_0 = c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + \cdots + c_n\mathbf{u}_n$$

Since $A^k\mathbf{u}_k = \lambda_i^k\mathbf{u}_k$ for each eigenvector $\mathbf{u}_i$ and each positive integer k, if we form the product $A^k\mathbf{x}_0$ we get

$$\begin{aligned} A^k\mathbf{x}_0 &= A^k(c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + \cdots + c_n\mathbf{u}_n) \\ &= c_1\lambda_1^k\mathbf{u}_1 + c_2\lambda_2^k\mathbf{u}_2 + \cdots + c_n\lambda_n^k\mathbf{u}_n \end{aligned}$$

Next, divide both sides by λ_1^k. (This is similar to our scaling in each step of the Power Method.) This gives us

$$\left(\frac{1}{\lambda_1}\right)^k A^k\mathbf{x}_0 = c_1\mathbf{u}_1 + \left(\frac{\lambda_2}{\lambda_1}\right)^k\mathbf{u}_2 + \cdots + \left(\frac{\lambda_n}{\lambda_1}\right)^k\mathbf{u}_n$$

By (5), as k gets large, each of $\left(\frac{\lambda_2}{\lambda_1}\right)^k, \cdots, \left(\frac{\lambda_n}{\lambda_1}\right)^k$ gets smaller. Hence as $k \to \infty$,

$$\left(\frac{1}{\lambda_1}\right)^k A^k\mathbf{x}_0 \to c_1\mathbf{u}_1$$

as claimed and observed in our examples.

Note that the larger $|\lambda_2|$ is relative to $|\lambda_2|, \ldots, |\lambda_n|$, the faster $\left(\frac{\lambda_2}{\lambda_1}\right)^k, \cdots, \left(\frac{\lambda_n}{\lambda_1}\right)^k$ converge to 0. This is why the Power Method converges more rapidly when the dominant eigenvalue is much larger than the other eigenvalues.